

EDUCATION AND ACADEMIC POSITIONS

Independent Contractor, Apple MLR

Working on AI reasoning for formal theorem proving.

Lausanne, Switzerland

Sep. 2025–Sep. 2027

PostDoctoral Researcher, MDS lab, EPFL

Lausanne, Switzerland

Sep. 2025–Sep. 2027

Postdoctoral Fellow at the ETH AI Center

Working on algorithms at the intersection of ML and Theory.

Zürich, Switzerland

Jul. 2023–Jul. 2025

École Polytechnique Fédérale de Lausanne (EPFL)

Ph.D. in Theoretical Computer Science, Advisor: Ola Svensson.

Lausanne, Switzerland

2018–2023

Université Paris Diderot - École Normale Supérieure Paris

M.Sc. in Theoretical Computer Science (with highest honors, ranked second).

Paris, France

2017–2018

– Thesis: “Local algorithms in graphs”, Advisor: Louis Esperet.

École polytechnique (Engineering diploma)

Specialization in Machine Learning and Algorithms.

Paris, France

2014–2018

Lycée Louis-Le-Grand, classe préparatoire MPSI/MP*

Training in mathematics and physics for competitive exams to enter “Grandes écoles”.

Paris, France

2012–2014

French “Baccalauréat série S” (with highest honors).

Paris, France

2012

INTERNSHIPS AND ACADEMIC VISITS

Academic visit at the Simons Institute

Visitor of the “Data-Driven Decision Processes” semester program.

Berkeley, USA

Nov. 2022–Dec. 2022

Academic visit at ETH Zürich

Visitor of Prof. Rico’s Zenklusen group

Zurich, Switzerland

May 2022–June 2022

Research internship G-SCOP

Research in the G-SCOP team for combinatorial optimization.

Grenoble, France

April 2018–August 2018

– I investigated what can or cannot be computed by local algorithms in graphs. This project led to two publications in theoretical computer science conferences.

Research and software engineering internship, INRIA

Research done in the INRIA team NANO-D.

Grenoble, France

March 2017–July 2017

– I designed and implemented an efficient algorithm for detecting symmetries in folded protein in order to predict their interactions. The programming language was C++.

Internship at Surrey Satellite Technology Ltd

Intern in the OBDH (On-Board Data Handling) team.

Guildford, England

Summer 2017

- I took part in analyzing circuit designs and optical devices for satellites as well as programming analysis software.

Military Service

Paris Fire Brigade, first fire group.

Paris, France

September 2014–April 2015

- Team leader of a rescue team of 3 people in case of accident, dangerous disease, or fire.

FUNDING AND RECOGNITION

• ETH AI Center Postdoctoral fellowship	2023-2025
• Grant from the Bernoulli Center to organize the ALPS  workshop (~30000 chf)	2022
• NeurIPS oral presentation (Top 1% of submissions)	2020
• NeurIPS spotlight presentation (Top 4% of submissions)	2020
• EPFL IC School 1-year Fellowship	2018

PUBLICATIONS

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1. Kirill Brilliantov, Étienne Bamas, Emmanuel Abbé, “ k -server-bench: Automating Potential Discovery for the k -Server Conjecture” , 3rd *AI for Math workshop*, AI4Math@ICML 2026.
 2. Étienne Bamas, Shi Li, Lars Rohwedder, “Randomized Rounding over Dynamic Programs” , 58th *ACM Annual ACM Symposium on Theory of Computing*, STOC 2026.
 3. Étienne Bamas, Lars Rohwedder, Sarah Morell, “The Submodular Santa Claus Problem”, 36th *ACM-SIAM Symposium on Discrete Algorithms*, SODA 2025.
 4. Étienne Bamas, “Lift-and-Project Integrality Gaps for Santa Claus” , 36th *ACM-SIAM Symposium on Discrete Algorithms*, SODA 2025.
 5. Étienne Bamas, Sai Ganesh Nagarajan, Ola Svensson, “An Analysis of D^α seeding for k -means” , 41st *International Conference in Machine Learning*, ICML 2024.
 6. Étienne Bamas, Alexander Lindermayr, Nicole Megow, Lars Rohwedder, Jens Schölter, “Santa Claus meets Makespan and Matroids: Algorithms and Reduction” , 35th *ACM-SIAM Symposium on Discrete Algorithms*, SODA 2024.
Invited to the ACM Transactions on Algorithms special issue of SODA (top 5-10% of accepted papers).
 7. Étienne Bamas, Lars Rohwedder, “Better Trees for Santa Claus” , 55th *Annual ACM Symposium on Theory of Computing*, STOC 2023.
 8. Étienne Bamas, Marina Drygala, and Ola Svensson, “A Simple LP-Based Approximation Algorithm for the Matching Augmentation Problem” , 23rd *Conference on Integer Programming and Combinatorial Optimization*, IPCO 2022.
 9. Étienne Bamas, Marina Drygala, and Andreas Maggiori, “An Improved Analysis of Greedy for Online Steiner Forest” , 33rd *ACM-SIAM Symposium on Discrete Algorithms*, SODA 2022.
 10. Étienne Bamas, Paritosh Garg, and Lars Rohwedder, “The Submodular Santa Claus Problem in the Restricted Assignment Case” , 48th *International Colloquium on Automata, Languages, and Programming*, ICALP 2021.
 11. Étienne Bamas, Andreas Maggiori, and Ola Svensson, “The Primal-Dual method for Learning Augmented Algorithms” , *Advances in Neural Information Processing Systems 33*, NeurIPS 2020 (**oral presentation**).
 12. Étienne Bamas, Andreas Maggiori, Lars Rohwedder, and Ola Svensson, “Learning Augmented Energy Minimization via Speed Scaling” , *Advances in Neural Information Processing Systems 33*, NeurIPS 2020 (**spotlight presentation**).

13. Étienne Bamas and Louis Esperet, "Local Approximation of the Maximum Cut in Regular Graphs" [↗](#), *Graph-Theoretic Concepts in Computer Science - 45th International Workshop*, WG 2019.
14. Étienne Bamas and Louis Esperet, "Distributed Coloring of Graphs with an Optimal Number of Colors" [↗](#), *36th International Symposium on Theoretical Aspects of Computer Science*, STACS 2019.

TEACHING

- Lecturer for "AI Center Projects in Machine Learning Research" Spring 2024
- Lecturer for "Advanced Topics in Discrete Optimization" Spring 2024
- Supervision of master thesis at ETHZ since 2023
Supervised students: Gaia Torresani (Fall 2024), Thibault Vignon (Fall 2023), Jonathan Schnell (Spring 2024).
- Supervision of semester projects at EPFL 2021-2023
Supervised students: Alexandre Reynaud (master student), Taha El Ghazi (master student).
- Teaching Assistant at EPFL 2019-2022
"Algorithms" (head TA), "Theory of Computation" (head TA), "Information, Calcul, Communication"
- Teaching Assistant at Lycée Janson-de-Sailly 2017-2018
In charge of oral exams in mathematics

SKILLS AND LANGUAGES

- **Programming:** C++, Python, Pytorch, Java, Lean4, Numpy, Scipy.
- **French:** Native.
- **English:** Full professional proficiency (110/120 TOEFL iBT).
- **Spanish and Mandarin:** Basics.

ACADEMIC SERVICE AND TALKS

- I co-organized the workshop ALPS '22 [↗](#) on algorithms with predictions (~ 45 international participants).
- Program committee: MAPSP '24
- Reviewer for the conferences: MFCS '19, WAOA '20, ITCS '20, '23, NeurIPS '21, '22, '23, SODA '22, '23, ICML '22, STOC '23, '25, IPCO '24, '25, ICALP '26, EMNLP '26
- Reviewer for the journals: Algorithmica (2021), Journal of Computer and System Sciences (2023).
- I gave a talk at the following conferences or workshops: WG '19, STACS '19, Dagstuhl scheduling seminar in 2020 and 2023, NeurIPS '20, ICALP '21, Operations Research Bern (OR) '21, SODA '22, STOC '23, ICML '24.

EXTRACURRICULAR ACTIVITIES

- Fond of sports: running (find me on Strava [↗](#)), skiing, rock climbing, hiking.
- Member of EPFL emergency medical team (2019).